Surya Teja Anupindi

Machine Learning Engineer

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Machine Learning Engineer with a Master's in Data Science and hands-on experience building end-to-end ML pipelines across healthcare, environmental science, logistics, and agentic AI systems. Proficient in Python, PyTorch, TensorFlow, Scikit-learn, and vector databases with deep expertise in NLP, Computer Vision, stateful multi-agent system orchestration, and sandboxed sandbox execution configurations.

RELEVANT WORK EXPERIENCE

Quantela **Apr 2022 – May 2024**

Machine Learning Engineer Hyderabad, TG

- Architected end-to-end MLOps pipelines (ingestion -> training -> deployment -> monitoring) on AWS/GCP, reducing model deployment time from 6 weeks to under 48 hours and cutting batch training cost by ~45%.
- Pioneered an LLM-powered citizen-service automation workflow integrated into the eGovernance stack, resolving 89% of 40K+ monthly queries without human intervention and reducing response time from 14 min to under 30 sec.
- Established model governance (drift detection, SHAP-based explainability, A/B testing) aligned to CMMI Level 5, passing external AI audits with zero critical findings.

Quantela **2021 – Apr 2022**

Machine Learning Intern Hyderabad, TG

- Productionized computer vision and anomaly-detection models for video monitoring, crowd management, and IoT telemetry across 35,000+ smart-streetlight and parking assets, achieving 94% mAP and a 62% drop in false-positive alerts.
- Built a reusable AI model store and feature-engineering framework that eliminated ~70% of duplicate modeling effort across Asset Management, Location Insights, and Citizen App product lines.

PROJECTS

ArgRAG: Argument-Aware Retrieval-Augmented Generation **2025**

Research Paper

- Improved factual accuracy by 3.4% and explanation quality by 21% (human-rated) by designing ArgRAG, a modular RAG pipeline that inserts an explicit argumentation layer between retrieval and generation for explainable NLP.
- Reduced retrieval variance by ensembling BM25, DPR, ColBERT, and SPLADE retrievers via Reciprocal Rank Fusion, achieving +4.4% Verdict EM over the best single retriever on the BEIR-FEVER benchmark.

Autonomous Software Engineering & Agentic Serving Platform **2026**

Python, LangGraph, crewAI, AutoGen, vLLM, PEFT/LoRA, Docker, Pytest

- Designed and implemented a self-correcting autonomous coding loop in Python that executes test suites via pytest subprocesses inside isolated Docker containers, capturing stack traces and patching buggy scripts automatically.
- Modeled and compiled stateful multi-agent workflows using LangGraph and crewAI to coordinate Programmer, Code Reviewer, and Structured Output Payload Router agent configurations.
- Set up high-performance model serving using vLLM to host Mistral-7B at 4-bit AWQ precision; built a streaming client to measure Time-to-First-Token (TTFT) latency and token throughput (tokens/sec).

PROJECTS (CONTINUED)

Multimodal Clinical Outcome Prediction – MIMIC-IV **2025**

Python, PyTorch, BioClinicalBERT, SQLAlchemy

- Improved patient outcome F1 score by 7% by building a multimodal deep learning model integrating structured EHR data and BioClinicalBERT clinical note embeddings, benchmarking baseline model architectures.

Anomaly Detection in Chest Radiographs **2024**

Python, TensorFlow, Keras, OpenCV, Scikit-learn

- Achieved reliable pneumonia and tuberculosis detection using a CNN-based pipeline with image preprocessing/augmentation, validating predictions via ROC curves and Grad-CAM visualizations.

Species Distribution & Environmental Factor Analysis **2024**

Python, Pandas, Scikit-learn, Matplotlib

- Identified biodiversity drivers by analyzing species distribution against environmental variables using IQR-based outlier detection, benchmarking Linear Regression, Random Forest, and Gradient Boosting.

Asteroid Diameter Prediction via MLP Neural Network **2024**

Python, TensorFlow, Scikit-learn, Pandas

- Achieved reliable asteroid size estimation by optimizing an MLP neural network on NASA's open dataset, delivering strong regression performance (MAE, MSE, R^2).

Delivery Route Optimizer with Real-Time Constraints **2024**

Python, NetworkX, NumPy, Pandas

- Developed a real-time route optimization system using Dijkstra, A*, and TSP-based heuristics, validating scalability across distance, latency, and constraint satisfaction metrics.

EDUCATION

Saint Peter's University **Feb 2026**

Master of Science in Data Science Jersey City, NJ

Gandhi Institute of Technology and Management **Aug 2023**

Bachelor of Technology in Computer Science & Engineering India

SKILLS

Technical Skills: Deep Learning (PyTorch, TensorFlow, Keras), NLP, Transformers (Hugging Face, BERT, RoBERTa, DeBERTa, BioClinicalBERT, FLAN-T5), RAG & IR (BM25, DPR, ColBERT, SPLADE, FAISS, RRF), Agentic AI (LangGraph, crewAI, AutoGen), LLM Serving & Optimization (vLLM, PEFT/LoRA, AWQ, TTFT), Computer Vision (OpenCV, CNNs, Grad-CAM, anomaly-detection), MLOps, Sandbox Security & Testing (Docker, Pytest)

Languages & Tools: Python, SQL, R, Java, C, SQLAlchemy, Git, Anaconda, NLTK, SacreBLEU, Unix/Linux, Bash scripting